AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 33 · STUDENT LESSON

Independent reproduction check

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall Peer review and response. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of independent reproduction check. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of independent reproduction check.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Peer review and response. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Writing the methods section. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Robustness across conditions. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of independent reproduction check

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Peer review and response. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain independent reproduction check using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Independent reproduction check case. Separate observations, given facts and assumptions before choosing an explanation.

Independent reproduction checks whether another operator can use the recorded artifacts to recover the stated computation without hidden help.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Independent reproduction check. Connect each labelled input, deciding step and result.

Provide manifest, fixture data, code, environment and expected metric definition. Record assistance and mismatches; reproduction is not new-data replication.

Independent reproduction check — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Operator gets different MSE because the supplied script averages absolute errors while paper names squared error.	Provide manifest, fixture data, code, environment and expected metric definition. Record assistance and mismatches; reproduction is not new-data replication.	Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Independent reproduction check. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Operator gets different MSE because the supplied script averages absolute errors while paper names squared error. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Q3's saved file is missing from the archive.

Predicted result

Prediction: Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed.

Why the result follows

Reason: A missing record prevents checking the contested decision.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Independent reproduction check to explain a concrete decision, rather

> than only memorising definitions.

Word	Meaning
Independent operator	A person executing from the supplied record
Reproduction discrepancy	A mismatch needing documented investigation
Artifact manifest	A versioned inventory for the reported computation

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Independent reproduction check, predict a step, then compare with its explanation.

Operator gets different MSE because the supplied script averages absolute errors while paper names squared error.

Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

Worked explanation · Independent reproduction check

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

> The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Independent reproduction check

An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score.

The facts we will use

> **Supplied facts**
>
> Original correct count is 5/6. Rerun is 4/6. Case Q3 is the only scoring disagreement. No new model **output**s are generated.

AI ACADEMY / AI-7/33

Independent reproduction check

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compare exact artefacts

Verify data and rubric versions before changing the result.



2. Locate disagreement

The score difference is concentrated in Q3.



3. Resolve by the rubric

Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

CHANGE ONE THING

Q3's saved file is missing from the archive.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compare exact artefacts	Verify data and rubric versions before changing the result.
2. Locate disagreement	The score difference is concentrated in Q3.
3. Resolve by the rubric	Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

An independent reproduction check can uncover interpretation or implementation differences. Do not average incompatible scores or simply choose the higher one.

What this example does not establish

Keep the stated limit

This check concerns saved **output**s and scoring, not new-sample replication or future model behaviour.

Keep the stated limit

Read the labelled picture: Locate the Q3 scoring disagreement between 5/6 and 4/6. What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?

Keep the stated limit

Change one thing: Q3's saved file is missing from the archive. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Independent reproduction check claim and explain a repair using the supplied evidence.

A successful reproduction repeats the original **evidence** and may preserve its original design limitations.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: A successful reproduction repeats the original **evidence** and may preserve its original design limitations.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Independent reproduction check with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Independent reproduction check”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Independent reproduction check. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Locate the Q3 scoring disagreement between 5/6 and 4/6. What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Independent reproduction check. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Operator gets different MSE because the supplied script averages absolute errors while paper names squared error. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Independent reproduction check. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: A successful reproduction repeats the original **evidence** and may preserve its original design limitations.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final **evidence** pack: Reproducer uses different validation split because no split IDs were provided. Diagnose.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Independent reproduction check case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Independent reproduction check — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Operator gets different MSE because the supplied script averages absolute errors while paper names squared error.	Provide manifest, fixture data, code, environment and expected metric definition. Record assistance and mismatches; reproduction is not new-data replication.	Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Reproducer uses different validation split because no split IDs were provided. Diagnose.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Reproducer uses different validation split because no split IDs were provided. Diagnose.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Independent reproduction check, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score.

By the end, I can	How I will show it
Explain the mechanism and terminology of independent reproduction check	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Independent reproduction check without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of independent reproduction check

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

1. Resolve by the rubric

2. Compare exact artefacts

3. Locate disagreement

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Resolve by the rubric Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

An independent reproduction check can uncover interpretation or implementation differences. Do not average incompatible scores or simply choose the higher one.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Q3's saved file is missing from the archive. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed. A missing record prevents checking the contested decision.

Repair a missing building block

Review Peer review and response (ai-7/32). Goal: Explain the **mechanism** and terminology of peer review and response. Short practice: Count photos and underlying plants in the reviewer's concern.

What analysis assumption needs revision, and why does changing only the word photos to plants fail to repair it?. Return to this week and explain what you changed.

Review Writing the methods section (ai-7/30). Goal: Explain the **mechanism** and terminology of writing the methods section. Short practice: Apply the inclusive four-character rule to oak, pine and willow.

Which label changes under greater-than-four, and what method detail allows another team to reproduce the distinction?. Return to this week and explain what you changed.

Review Robustness across conditions (ai-7/27). Goal: Explain the **mechanism** and terminology of robustness across conditions. Short practice: Calculate indoor, outdoor and combined rates.

Which relevant weakness disappears from view when only the combined rate is reported?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

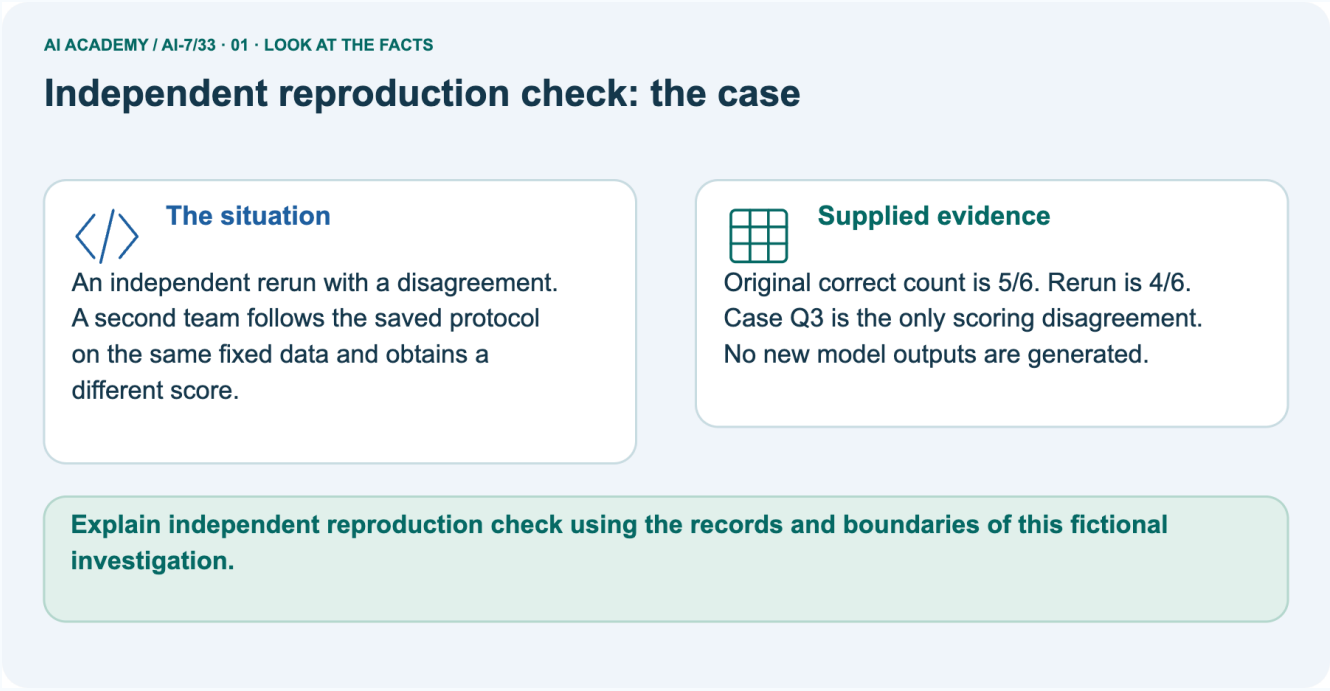
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Independent reproduction check graphic using its labels, deciding evidence and relevant limitation.

01. Independent reproduction check: the case



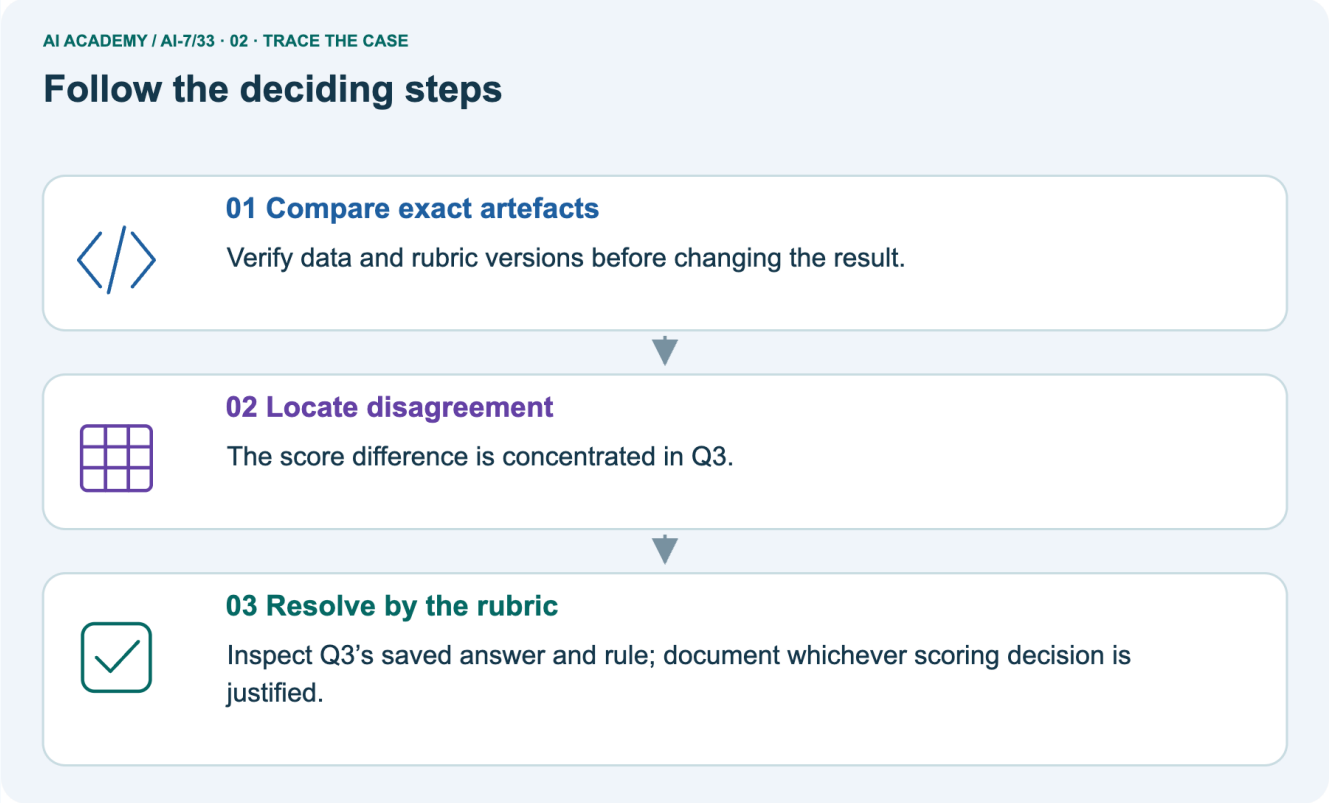
Follow the labelled picture with the matching explanation. Open the vector diagram

An independent rerun with a disagreement. A second team follows the saved protocol on the same fixed data and obtains a different score. Supplied facts: Original correct count is 5/6. Rerun is 4/6.

Case Q3 is the only scoring disagreement. No new model **output**s are generated.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

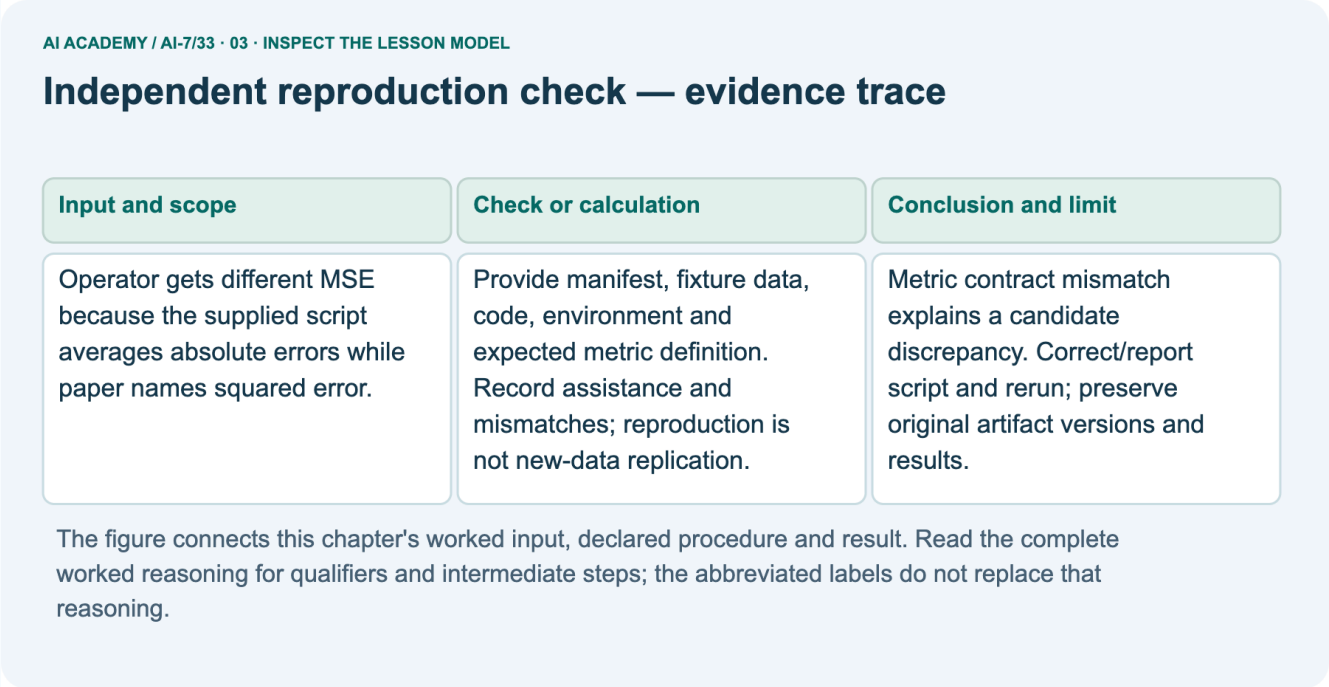


Follow the labelled picture with the matching explanation. Open the vector diagram

Compare exact artefacts: Verify data and rubric versions before changing the result. → Locate disagreement: The score difference is concentrated in Q3. → Resolve by the rubric: Inspect Q3's saved answer and rule; document whichever scoring decision is justified.

Point to each numbered stage and explain which recorded **input** it uses.

03. Independent reproduction check — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

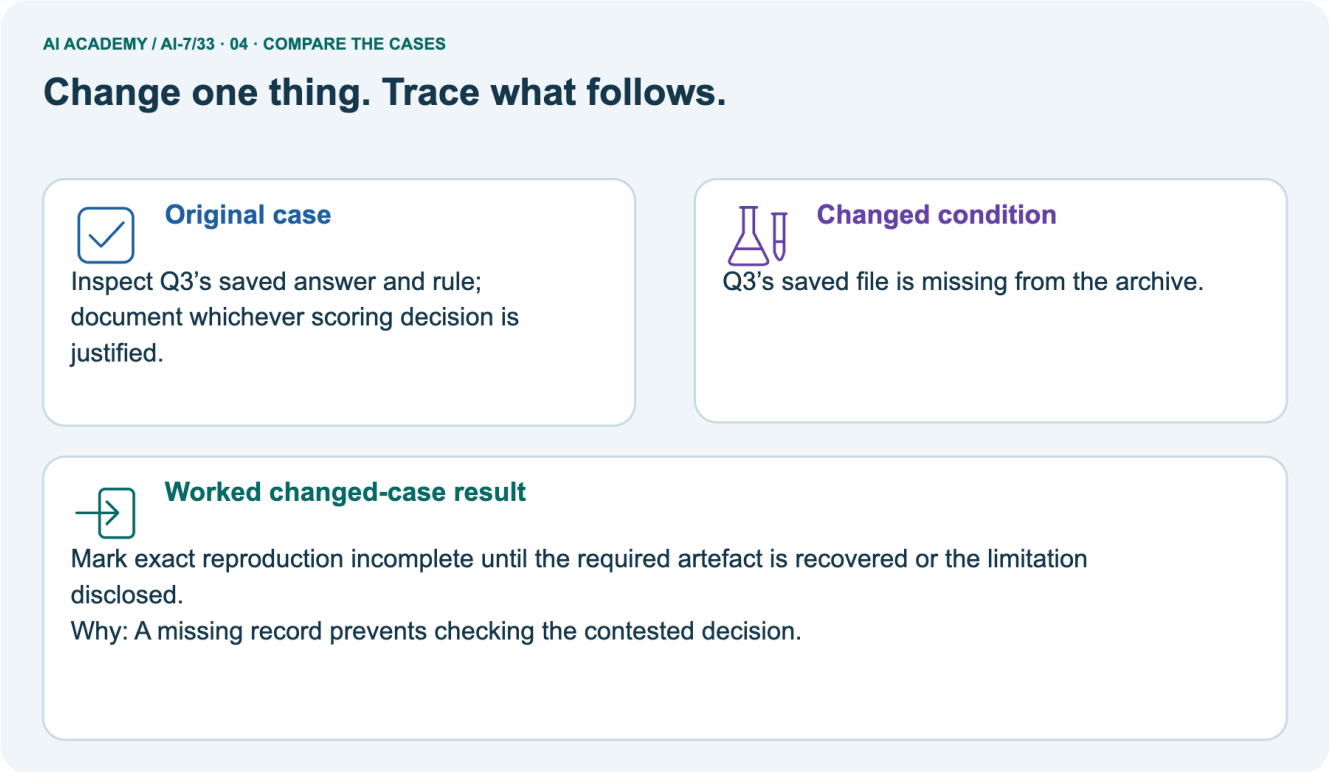
Independent reproduction check — **evidence** trace.

Operator gets different MSE because the supplied script averages absolute errors while paper names squared error. | Provide manifest, fixture data, code, environment and expected metric definition.

Record assistance and mismatches; reproduction is not new-data replication. | Metric contract mismatch explains a candidate discrepancy. Correct/report script and rerun; preserve original artifact versions and results.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



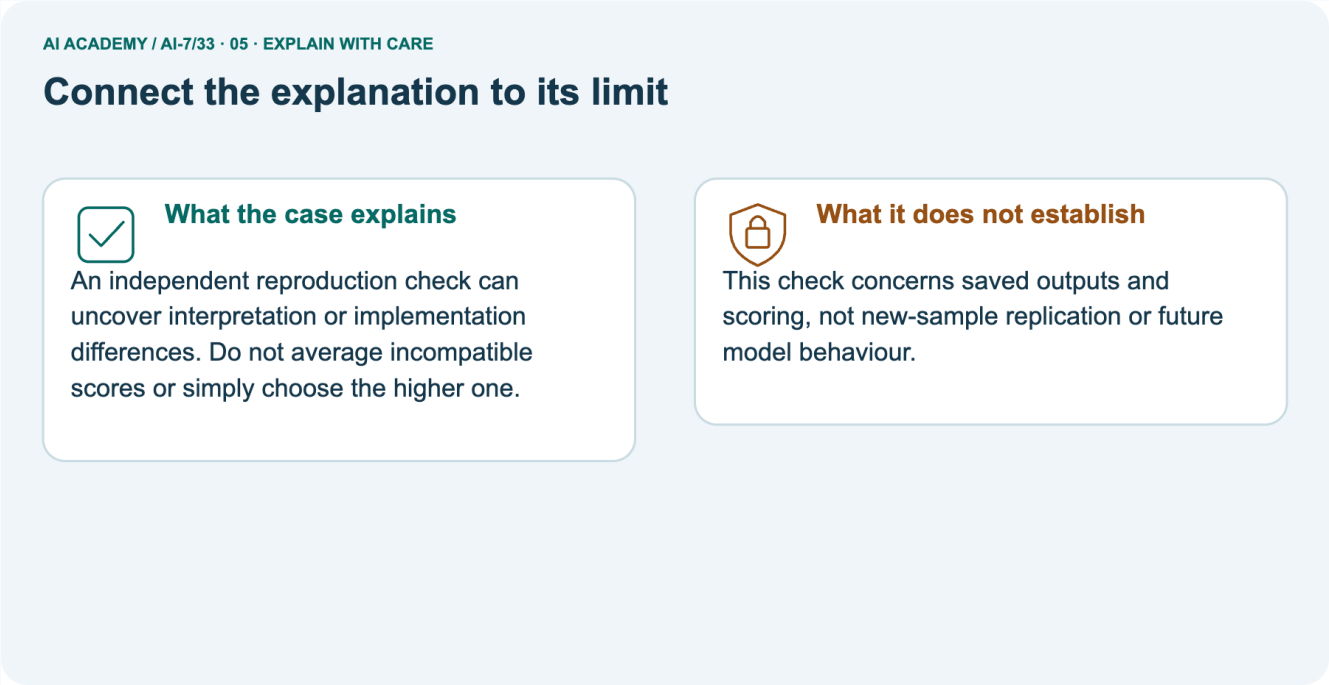
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Inspect Q3's saved answer and rule; document whichever scoring decision is justified. Changed condition: Q3's saved file is missing from the archive.

Expected result: Mark exact reproduction incomplete until the required artefact is recovered or the limitation disclosed. Reason: A missing record prevents checking the contested decision.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



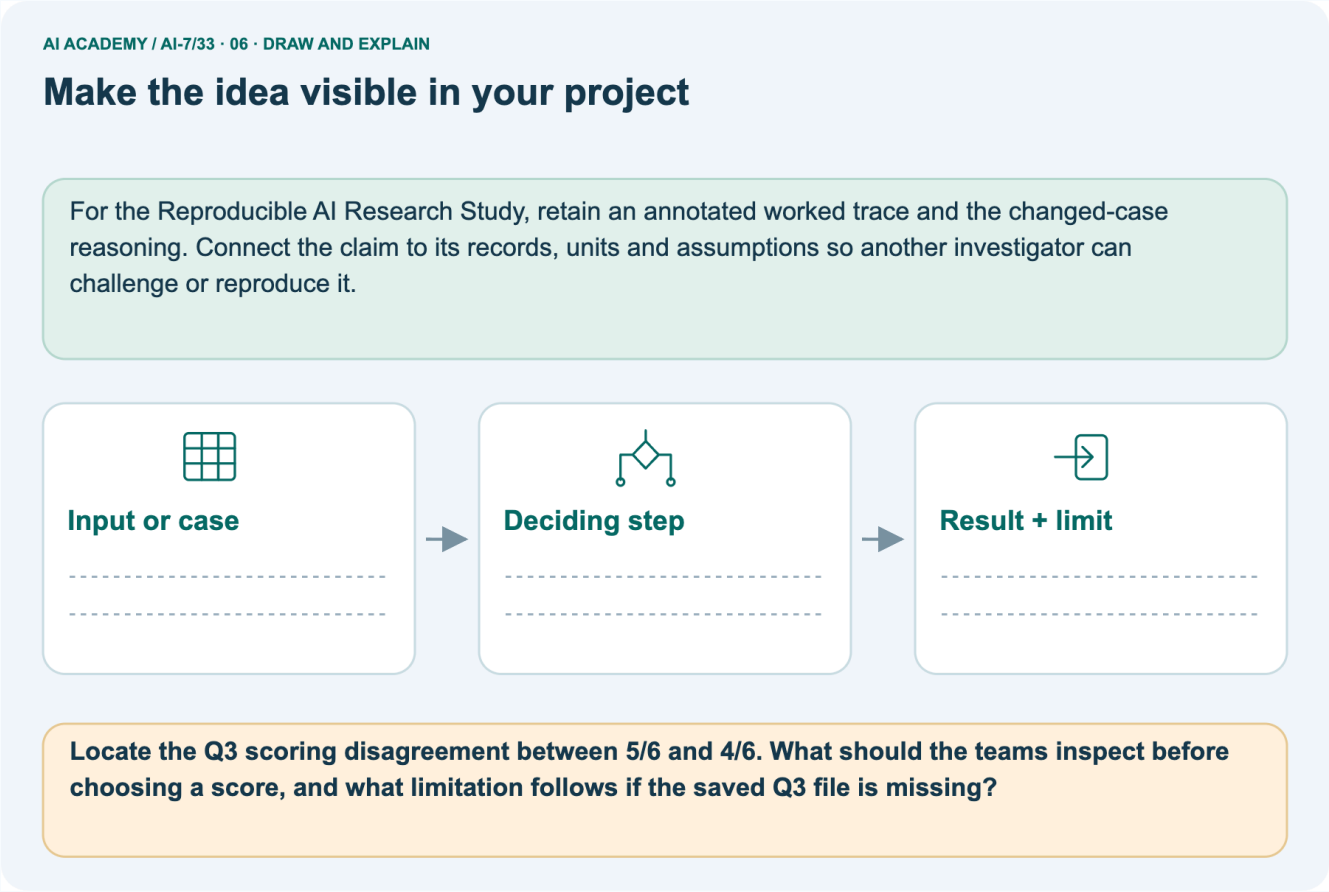
Follow the labelled picture with the matching explanation. Open the vector diagram

An independent reproduction check can uncover interpretation or implementation differences. Do not average incompatible scores or simply choose the higher one.

Limit: This check concerns saved **output**s and scoring, not new-sample replication or future model behaviour.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Locate the Q3 scoring disagreement between 5/6 and 4/6.

What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.